

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458321.7916 +/- 0.0053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.123 +/- 0.045
Transit depth (Rp/Rs)^2	1.51 +/- 1.11 [%]
Orbital Inclination (inc)	81.93 +/- 2.94
Airmass coefficient 1 (a1)	2.25 +/- 1.89
Airmass coefficient 2 (a2)	-0.52 +/- 1.02
Scatter in the residuals of the lightcurve fit is	104.24 %
Best Comparison Star	#2 - [302, 370]
Optimal Aperture	3.57
Optimal Annulus	3.71
Transit Duration (day)	0.028 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.123 ± 0.045 vs 0.1326 (deviation 0.14σ)
Duration fit vs published	0.028 d vs 0.0737 d (deviation 0.89σ)
Mid-time O–C	8.1 min
Depth SNR	1.8
Residual scatter	104.24 %

Light curve

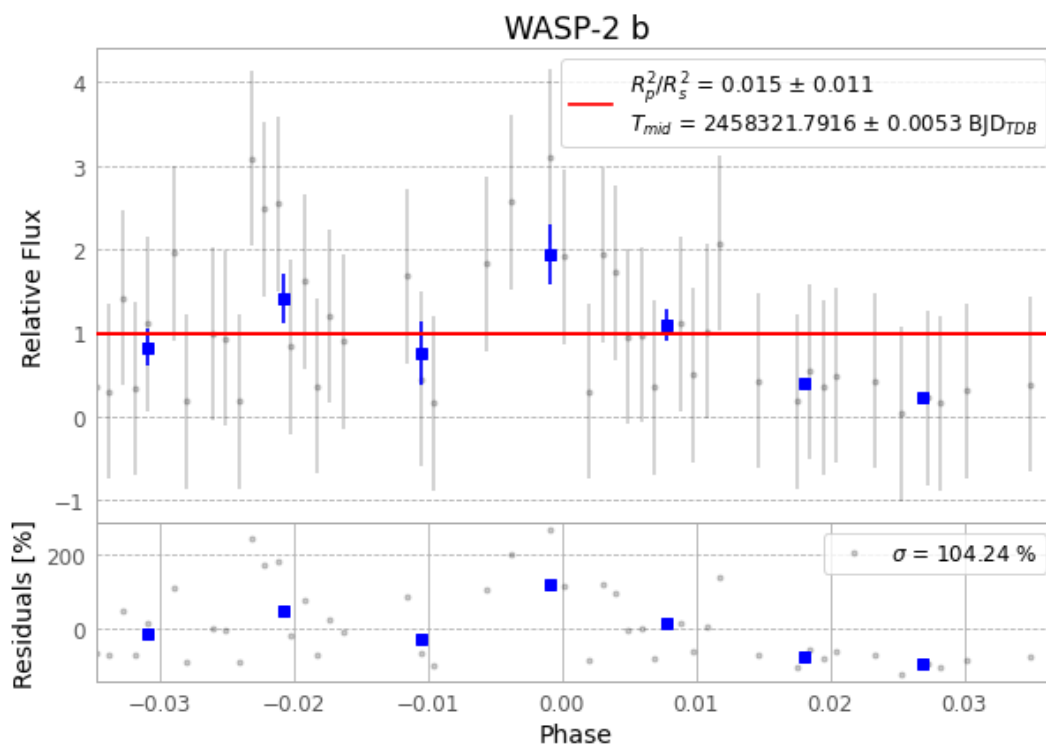


Figure 1 — FinalLightCurve_WASP-2 b_2018-07-21.png (SHA-256 3a3151261b218a77...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [457, 127], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 49d42c58be00bdb8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e

Data provenance

77 FITS frames (51 MB), WASP-2180722045712.FITS → WASP-2180722084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-180722.log (1c9dba8b2775...), FinalLightCurve_WASP-2 b_2018-07-21.png (3a3151261b21...), FinalLightCurve_WASP-2 b_2018-07-21.csv (087c9a30bb5e...)

Compute resources

Wall time 240.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 23:07Z.